



Recharge via twitter

Tata Docomo's e-commerce initiative, Twittcom, allows its prepaid customers to recharge their account using #Hashtags <http://dgit.in/Kc4bnx>

Google domains

The company has applied for suffixes such as .google and .youtube to 'explore potential source of innovation on web'

Why SoC serious?

Dr. Sandeep Sibal of Qualcomm spoke to us on the future of the Snapdragon SoC ecosystem, the philosophy behind designing power efficient SoCs and more. An excerpt...

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Q Considering there is such a race amongst system-on-chip (SoC) manufacturers with so many smartphones and tablets sporting them, what is Qualcomm doing to stand out of the crowd? What would you want an end user to know about the advantages of the Snapdragon SoC vis-a-vis the competition? Competition is good for consumers and the industry and we at Qualcomm welcome competition. It forces everyone to innovate and bring out compelling products. We have established smartphone SoC leadership for Snapdragon processors by engineering excellence into all key parameters including CPU performance, graphics, connectivity and power efficiency.

Unlike many of our competitors, we design our own CPU micro-architecture from the ground up with mobile requirements in mind. Delivering the best mobile experience is all about combining leading technologies and integrating them into a single solution. This is what Qualcomm customers have come to expect from Snapdragon.

Q With the manufacturing processes reducing in size – S1-65nm, S2 and S3 – 45nm and S4 – 28nm, the power consumed does come down, but what are the other challenges faced and how do you overcome them?

Although reducing the size of the process technology used in manufacturing process does help in lowering the power consumption, significant additional power savings can be accrued by optimizing the architecture of various CPU cores on the chipset. For example, Snapdragon uses independent power and frequency controls for



Dr. Sandeep Sibal, Country Manager and Vice President – Business Development, Qualcomm India and South Asia

each of its cores. This helps us run them in accordance with the processing load and avoid any undesirable wastage as well as any thermal radiation issues.

Unique to Qualcomm, our dual-core and quad-core Snapdragon processors feature asynchronous technology that can throttle each core independently. This ensures the phone always uses the optimal amount of CPU power for the task at hand without draining battery life. This results in a speedy, responsive smartphone that lasts you the whole day.

Q Nowadays smartphones come with dual-core processors or even quad-core processors. But are there enough apps that make use of so many cores?

Currently, very few mobile applications are optimized to take advantage of quad-core CPUs. For that reason, we at Qualcomm focus on designing highly efficient and intelligent quad-core CPUs so that the Snapdragon processor is optimized to provide the best user experience rather than rushing to be first to market. By the time that Snapdragon-enabled devices featuring quad core CPUs are ready to come to market in the second half of 2012,

the mobile developer community will be ready to begin providing apps capable of harnessing the power of smartphones and tablets with these capabilities.

Q Could you elaborate on how you keep the power consumption lower in the Krait processor on the S4? It is said that in the Krait CPU, due to the use of asynchronous multi-processing, each CPU core has an individual clock. Will that not add power circuitry on the die?

As discussed above, asynchronous multi-processor architecture, ensures the phone always uses the optimal amount of CPU power for the task at hand without draining battery life.

For example, our Snapdragon APQ8064 is a system on chip that includes four Krait CPUs in addition to the Adreno GPU, dedicated video and audio processors and a host of other components. We then integrate all of these advanced technology parts into one fully integrated system on chip. We believe this integrated approach results in better overall performance at lower power. On a pure CPU level, clearly, with more CPU cores comes increased mobile processing performance. However, it is important to note that more CPU or GPU cores do not necessarily mean a better mobile device or user experience. Otherwise, more cores can drain battery life at a very high rate. Qualcomm builds its Snapdragon processors from the ground up, tightly integrating the hardware and software. This ensures optimal power efficiency for both our dual-core and quad-core versions. It's about the user experience, not number of cores in the device.

Adding asynchronous multi-processing capability does add a little circuitry but it is insignificant when compared to the benefit it brings by reducing unwanted power consumption, and allowing each core to operate at optimal efficiency. It involves significantly less circuitry than adding an additional core as some of our competitors have done. Read more at: http://dgit.in/qualcomm_july2012